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EXPERIMENT 6

Aim: To study and implement infrastructure as a service using AWS.

Theory:

Amazon Web Services (AWS) offers a comprehensive Infrastructure as a Service (IaaS) platform, providing scalable and flexible cloud computing resources. Key offerings include **Amazon EC2** for virtual servers, **Amazon S3** for scalable storage, and **Amazon VPC** for networking, allowing users to securely manage and configure their virtual environments. AWS enables businesses to run a variety of workloads, from web hosting and big data processing to disaster recovery, all while offering pay-as-you-go pricing, high availability, and robust security features. With extensive management tools, like **CloudWatch** and **IAM**, AWS ensures users can monitor, automate, and secure their infrastructure effectively.

Steps:

1)

Instances (1)[Info](#)

Last updated less than a minute ago

Connect

Instance state ▾

Actions ▾

Launch instances ▾

Find Instance by attribute or tag (case-sensitive)

All states ▾

< 1 >

☐	Name ↕	Instance ID	Instance state ▾	Instance type ▾	Status check	Alarm status	Availability Zone ▾	Public IPv4 DNS ▾	Public IPv4 ... ▾	Elastic IP
☐	CSE(ICB)	i-09910654a4905e554	Running	t2.micro	2/2 checks passed View alarms +		ap-south-1b	ec2-65-2-35-245.ap-so...	65.2.35.245	-

2)



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Instances (1/1) Info

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
CSE(ICB)	i-09910654a4905e554	Running	t2.micro	2/2 checks passed		ap-south-1b	ec2-65-2-35-245.ap-so...	65.2.35.245	-

i-09910654a4905e554 (CSE(ICB))

Instance summary

Instance ID i-09910654a4905e554	Public IPv4 address 65.2.35.245 open address	Private IPv4 addresses 172.31.5.5
IPv6 address -	Instance state Running	Public IPv4 DNS ec2-65-2-35-245.ap-south-1.compute.amazonaws.com open address
Hostname type IP name: ip-172-31-5-5.ap-south-1.compute.internal	Private IP DNS name (IPv4 only) ip-172-31-5-5.ap-south-1.compute.internal	Elastic IP addresses -
Answer private resource DNS name IPv4 (A)	Instance type t2.micro	

3)

Instances (1/1) Info

Public IPv4 ...	Elastic IP	IPv6 IPs	Monitoring	Security group name	Key name	Launch time	Platform...	Managed	Operator
65.2.35.245	-	-	disabled	launch-wizard-1	CSE(ICB)	2025/02/18 10:41 GMT+5:30	Linux/UNIX	false	-

i-09910654a4905e554 (CSE(ICB))

Instance summary

Instance ID i-09910654a4905e554	Public IPv4 address 65.2.35.245 open address	Private IPv4 addresses 172.31.5.5
IPv6 address -	Instance state Running	Public IPv4 DNS ec2-65-2-35-245.ap-south-1.compute.amazonaws.com open address
Hostname type IP name: ip-172-31-5-5.ap-south-1.compute.internal	Private IP DNS name (IPv4 only) ip-172-31-5-5.ap-south-1.compute.internal	Elastic IP addresses -
Answer private resource DNS name IPv4 (A)	Instance type t2.micro	



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4)



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Preferences

Page size

☐ 10 resources

☐ 25 resources

☒ 50 resources

☐ Wrap lines
Enable to wrap table cell content, disable to truncate text.

☐ Striped rows
Select to add alternating shaded rows

☒ Use tags as suggestion options
Enables tags to be included in auto-suggestion options. (Decreases Performance if you have a large set of tags)

☒ Context menu
Replace the browser context menu on table rows to allow quick actions.

☒ Row click selection
Enable to allow clicking anywhere on a row item to select the item

☒ Compact table mode
Enabled compact table mode across the application making table rows more compact.

Attribute columns

Select visible attribute columns

- ☒ Instance ID (default)
- ☒ Instance state
- ☒ Instance type
- ☒ Status check
- ☒ Alarm status
- ☒ Availability Zone
- ☒ Public IPv4 DNS
- ☒ Public IPv4 address
- ☒ Elastic IP
- ☒ IPv6 IPs
- ☐ Private DNS name
- ☐ Private IP address
- ☒ Monitoring
- ☒ Security group name
- ☐ Security group IDs
- ☒ Key name
- ☐ Owner ID
- ☐ Volume ID

First column

Keep the first column(s) visible while horizontally scrolling the table content.

☒ None

☐ First column

☐ First two columns

Last column

Keep the last column visible while horizontally scrolling the table content.

☒ None

Cancel

Confirm



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Instance summary for i-09910654a4905e554 (CSE(ICB))

Updated less than a minute ago

Instance ID
i-09910654a4905e554

Public IPv4 address
65.2.35.245 | [open address](#)

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-5-5.ap-south-1.compute.internal

Instance type
t2.micro

VPC ID
vpc-0aee48773eb775848

Subnet ID
subnet-0669724789ee3b7f6

Instance ARN
arn:aws:ec2:ap-south-1:398302715217:instance/i-09910654a4905e554

Private IPv4 addresses
172.31.5.5

Public IPv4 DNS
ec2-65-2-35-245.ap-south-1.compute.amazonaws.com | [open address](#)

Elastic IP addresses
-

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | [Learn more](#)

Auto Scaling Group name
-

Managed
false

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance details

AMI ID
ami-00bb6a80f1f03502

AMI name
ubuntu/images/hvm-ssd-gp3/ubuntu-noble-24.04-amd64-server-20250115

Stop protection
Disabled

Monitoring
disabled

Allowed image
-

Launch time
Tue Feb 18 2025 10:41:21 GMT+0530 (India Standard Time) (6 minutes)

Platform details
Linux/UNIX

Termination protection
Disabled

AMI location
amazon/ubuntu/images/hvm-ssd-gp3/ubuntu-noble-24.04-amd64-server-20250115

6)

Stop instance

Stopping your instance allows you to reduce costs, modify settings, and troubleshoot problems.

Instance ID
i-09910654a4905e554 (CSE(ICB))

Stop protection
Off (Can stop instance)

You will be billed for associated resources
After you stop the instance, you are no longer charged usage or data transfer fees for it. However, you will still be billed for associated Elastic IP addresses and EBS volumes.

Associated resources
You will continue to incur charges for these resources while the instance is stopped

[Cancel](#) [Stop](#)

7)



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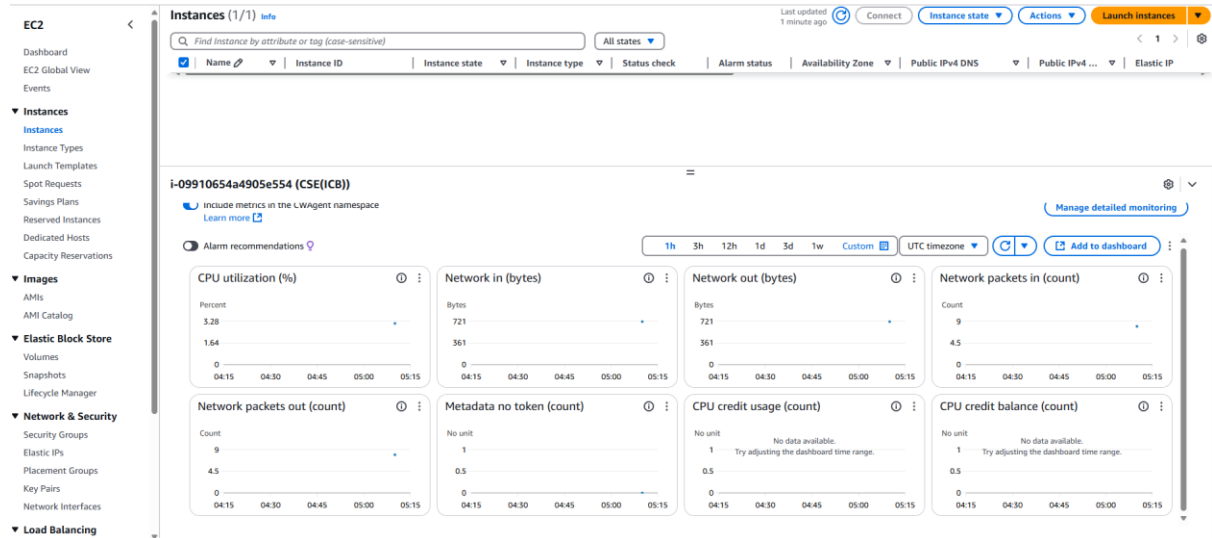


Name: Gargi Sukhatankar

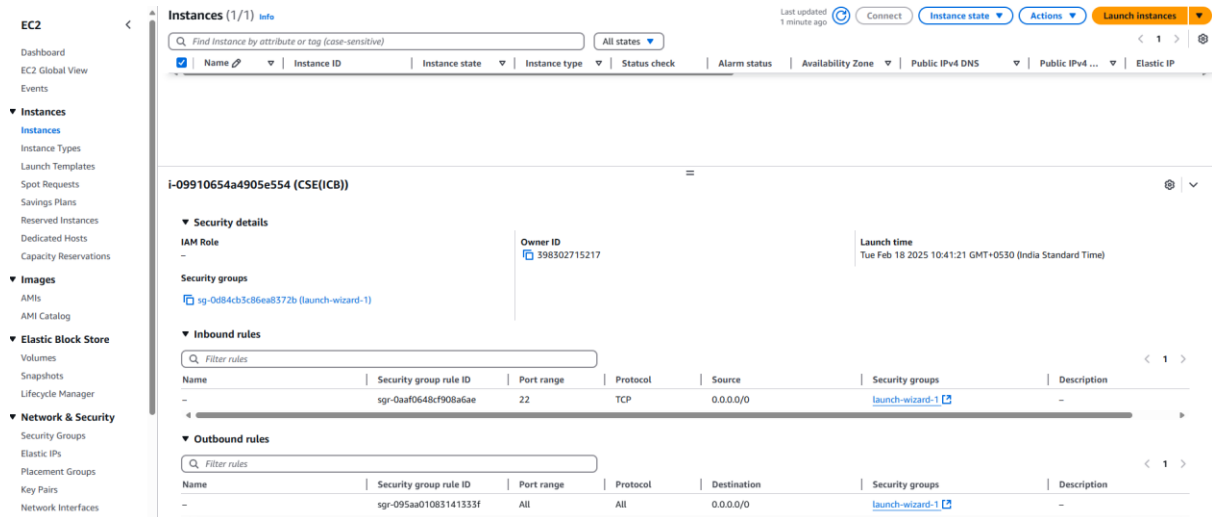
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8)



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EC2

- Dashboard
- EC2 Global View
- Events

▼ Instances

- Instances
- Instance Types
- Launch Templates
- Spot Requests
- Savings Plans
- Reserved Instances
- Dedicated Hosts
- Capacity Reservations

▼ Images

- AMIs
- AMI Catalog

▼ Elastic Block Store

- Volumes
- Snapshots
- Lifecycle Manager

▼ Network & Security

- Security Groups
- Elastic IPs
- Placement Groups
- Key Pairs
- Network Interfaces

i-09910654a4905e554 (CSE(ICB))

Details | Status and alarms | Monitoring | Security | **Networking** | Storage | Tags

▼ Networking details info

Public IPv4 address	Private IPv4 addresses	VPC ID
-	172.31.5.5	vpc-0a6c46f73eb773b48
Public IPv4 DNS	Private IP DNS name (IPv4 only)	Secondary private IPv4 addresses
-	ip-172-31-5-5.ap-south-1.compute.internal	-
Subnet ID	IPv6 addresses	Outpost ID
subnet-0669724789ee3b7fd	-	-
Availability zone	Carrier IP addresses (ephemeral)	
ap-south-1b	-	
Use RBN as guest OS hostname	Answer RBN DNS hostname IPv4	
Disabled	Enabled	

▼ Network Interfaces (1) info

Filter network interfaces

Interface ID	Device index	Card index	Description	Public IPv4 address	Private IPv4 address	Private IPv4 DNS	IPv6 addresses	Primary IPv6 address

10)

i-09910654a4905e554 (CSE(ICB))

Details | Status and alarms | Monitoring | Security | **Storage** | Networking | Tags

▼ Root device details

Root device name	Root device type	EBS optimization
/dev/sda1	EBS	disabled

▼ Block devices

Filter block devices

Volume ID	Device name	Volume size (GiB)	Volume State	Attachment status	Attachment time	Encrypted	KMS key ID
vol-0c2db63580ab0e660	/dev/sda1	8	In-use	Attached	2025/02/18 10:41 GMT+5:30	No	-

Volume monitoring (1)

1h 3h 12h 1d 3d 1w Custom UTC timezone Add to dashboard

Average read latency (ms/op)	Average write latency (ms/op)	Read throughput (KiB/s)	Write throughput (KiB/s)
No unit 0.979 0.49	No unit 1.89 0.943	No unit 807 403	No unit 582 291

Conclusion: Thus we successfully implemented infrastructure as a service using AWS.